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BSIT DA 2-1

**What is an algorithm, and how is it related to the tasks you perform in your daily life?**

An algorithm is a set of instructions that are different with each specific task. Algorithms are related to tasks performed daily as each thing we do has a set of steps or instructions to follow. For example, before eating you need to pick which food to cook, cook the food, eat the food, and clean the plates. This shows that even simple daily tasks work like algorithms, where following the right steps in the right order leads to the expected outcome.

**Why is it important for an IT student to learn algorithms?**

Algorithms are important for IT students because algorithms in coding are much more efficient than brute forcing a code. Algorithms are the foundation of problem-solving in programming. Before writing code, the person needs to think of the logical steps to solve a problem then they can figure out how to code their idea after. Algorithms can easily sort through data like the Binary search showcased in the linked video.

**1. for (let i = 1; i < 10; i *= 2) {**

**console.log(i);**

**}**

1. i = 1
2. if i < 10 continue to next step, otherwise stop
3. Print i
4. i = i * 2
5. Go back to step 2

**2. for (let i = 0; i < 10; i++) {**

**console.log(i);**

**}**

2. i = 0
3. If i < 10 continue to next step, otherwise stop
4. Print I
5. i = i + 1
6. Go back to step 2

**What did you observe about the sequence of numbers produced by the first code?**
The numbers are multiplied by two each time and stops stops once i is no longer less than 10.

**What did you observe about the sequence of numbers produced by the second code?**
The numbers start from 0 and increase by one each loop and stops stops once i is no longer less than 10.

**What is the main difference between the way the two loops progress?**
The 1st loop multiplies by 2 every loop, the 2nd loop only increases by one each loop.

**How does the way an algorithm performs its steps affect its output?**
The way algorithms perform may differ even with small changes to the code, in this case the only change was i = i++ and i = i * 2. This shows how small changes affect our code in big ways.